TOWARDS CONTENT-AGNOSTIC DEEPFAKE SPEECH DETECTION WITH MULTI-TTS

Thirulok Sundar Mohan Rasu[1], *Aswin Suresh*[2], *Arun Balajee Vasudevan*[3]

[1]University of Michigan, [2]DemoSquare, [3]Carnegie Mellon University

ABSTRACT

With the rise of advanced text-to-speech (TTS) systems and high-quality synthetic speech generation being increasingly accessible, reliably detecting fake speeches becomes ever more critical. Prior work on fake speech detection generally train models on real and fake speech datasets that differ in content, which inevitably result in the models using content-based features for distinguishing them. Such models are likely to fail to generalize to fake speeches that use content very different from those seen during training. In this work, we propose a content-agnostic deepfake speech detection model. We train this model on a dataset we curate of real–fake speech pairs where the *fake* speeches are synthesized using transcripts from real speech using three cutting-edge TTS systems—YourTTS, XTTS-v2, and OpenVoice. Through this approach we eliminate linguistic cues and force the model to rely solely on acoustic artifacts. Our proposed model has superior performance in comparison with prior work. Further, we study the effects of length of speech input on our model and on human performance, exploring both full-utterance speech and phrase-level variants. We also conduct extensive ablation studies and discuss several insights.

Index Terms— Fake Speech Detection, Content Agnostic Fake Speech

1. INTRODUCTION

The advances in Generative AI have enabled remarkable progress in content generation capabilities across language, vision, and audio modalities [1]. This quick progress comes with challenges such as "deep fakes", that lead to significant negative social impacts including spread of misinformation, impersonation, and reputation damage.

There have been significant efforts from the research community for developing fake speech detection algorithms [2, 3, 4]. Typically, such algorithms use neural network models trained on datasets of real and fake speeches. Majority of the works create real-fake speech pairs by manipulating real speeches. Real speech is transcribed to texts which can be further edited at word-level or phrase-level. This edited text is then converted to speech using Text to Speech (TTS) approaches as in AVDeepFake1M [4], FakeAVCeleb [5], VALL-E [6], Voicebox [7] and others. This approach has a crucial limitation that the models may learn to depend on the different content distribution of fake speeches, thereby losing generalization ability.

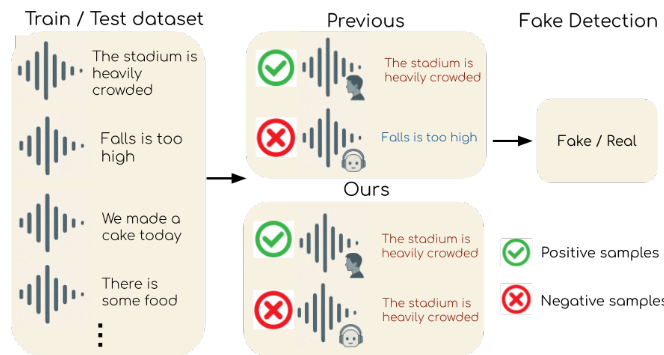

Fig. 1. Overview of our content-agnostic speech deepfake detection approach. Our model is trained using fake speeches with the same transcripts as real speeches, forcing the model to focus on audio features solely unlike previous works [4].

In contrast, our work focuses on learning content agnostic fake speech detectors. To create fake speech, we use a simple procedure of transcribing the real speech to text followed by using state-of-the-art TTS models [8, 9, 10] to generate the audio, without making any edits to the text in between. We thus ensure that the content in the real and fake speech are the same and only the audio characteristics are different. We show empirically that a discriminator trained to differentiate such real-fake pairs generalize better than content-aware fake speech detectors. There have been few previous attempts in a similar direction [2]. These works use codec-based TTS systems while we use more generalized TTS systems.

Further, we investigate the fake detection performance of humans and our learned model for different lengths of speech, gaining several interesting insights. Our proposed fake detection model's performance improves with increase in length of the speech sample. This trend is observe in humans as well. Over all lengths, our proposed model achieved better accuracy in comparison with humans.

Our main contributions are as follows: a) a content-agnostic fake speech detection benchmark curating a real-fake paired speech dataset comprising of $1M$ samples, b) a content-agnostic Fake Speech Detection (FSD) model with a detailed study varying the length of speech input, and c) study on human vs model performance on fake detection and their insights. In the next sections, we will detail related works (Section 2), our dataset curation and our high-level methodology (Section 3) and lastly our experimental results and discussions in Section 4 and 5.

2. RELATION TO PRIOR WORK

Deepfake Speech Detection. This mainly counters the voice spoofing synthesized by texts-to-speech (TTS) or voice conversion (VC) techniques or Physical access attacks [11]. Some works [12, 13, 14] apply pretrained speech representation models, such as wav2vec2 [15], to detect spoofed speech. Other works leverage classic speech tasks to improve speech deepfake detection; for example, DAD-SV [16] boils down deepfake detection into a speaker verification problem. Overall, extensive research has been dedicated to detecting deep fake detection in speech content. Primarily, we can view fake speech detection from two perspectives: content-aware and content agnostic. Let us address them below.

Content-aware Fake Speech Detection Content-aware approaches [4, 17, 18] for fake speech detection use transcribed or predicted linguistic content to detect inconsistencies between what is said and how it is said. Few approaches utilize phoneme-level modeling [19, 20] where the phoneme-level analysis enables fine-grained detection by examining inconsistencies in individual phonetic units rather than analyzing entire speech holistically [20]. Others incorporate end-to-end joint models that fuse ASR outputs with acoustic embeddings [21]. Ye *et al.*[22] proposed sematic decoupling framework for fake detection that learn synthesis-independent representations and sensitivity to manipulation artifacts across different generation techniques. Newly released AVDeepFake1M [4] fake detection benchmark compiles an audio-visual dataset where fakes are curated by semantically and seamlessly editing the real speech transcripts. This inevitably result in the models using content-based features for distinguishing *real / fake* samples.

Content-agnostic Fake Speech Detection Classical fake speech detection methods [23, 24] primarily rely on acoustic features like MFCCs [25], without considering the actual linguistic content of the speech. RawNet and its variants [26, 27] have demonstrated effectiveness in capturing subtle artifacts introduced by TTS and voice conversion systems. The ASVspoof challenges [28] have further standardized benchmarks and datasets for evaluating these content-agnostic systems. There are few neural network based approaches [2, 29, 30, 31]. The AASIST model represents a significant advancement, utilizing spectro-temporal graph attention mechanisms to capture spoofing artifacts [30]. Time Delay Neural Networks (TDNN) and their variants [29, 32] have demonstrated effectiveness in modeling long-term temporal dependencies crucial for spoofing detection.

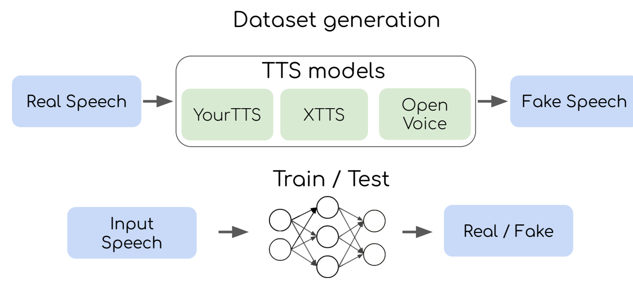

Fig. 2. Overview of our dataset creation method on the top row. Real speech (for accurate voice cloning) and its transcripts are given to the three s-o-t-a TTS systems to generate fake speech. Thus, we create real-fake paired speech with same content (transcripts). Bottom row depicts train/test pipeline. We provide input speech to our proposed fake detection model for binary classification of *real / fake*.

3. DATASET AND APPROACH

Our work focuses on fake speech detection. For this, we create a content-agnostic real-fake paired speech dataset, which comprises more than $1M$ speeches from more than 2000 subjects. We extract the clean speech of the speaker and the associated ground-truth transcript (from the metadata provided) from each real video of AV-DeepFake1M [4] dataset. Using transcript and speaker voice as a reference, we synthesize fake speech with three TTS systems: YourTTS [9], XTTS-v2 [10], and OpenVoice [8]. This way, we have paired real and fake speech from each TTS model. In total, we create approximately $250k$ real-fake pairs of full-length speech per model, with an average duration of about 15 seconds per audio. For further ablation studies, we also split the OpenVoice-generated real-fake paired speech into phrase-level speech segments. This results in separate dataset splits for each phrase length. We generate phrase-level data for around $100k$ audios for each segment length. Each full-length audio was split into at most 5 phrases for the phrase-level experiments.

Our approach is designed to establish a systematic and content-agnostic framework for fake speech detection. Prior work on fake speech detection generally train models on independent real and fake speech datasets, which inevitably result in the models using content-based features for distinguishing them. Such models are likely to fail to generalize to fake speeches using content very different from those seen during training. Unlike them, we maintain original transcripts for both real and fake audio, ensuring that the trained detection task depends solely on acoustic features and signal-level characteristics.

For model training, we employ an audio encoder and extend it with a two-layer MLP classification head to perform binary real-vs-fake detection task. We train the above model on our training data from different TTS systems as mentioned above as well as a combined model across all TTS sources

(YourTTS, XTTS-v2, and OpenVoice).

Furthermore, to analyze the effect of length of speech duration on model performance, we further segment TTS-generated full-length fake speech utterances into phrase-level speech clips containing 1-, 2-, 4-, and 8-word phrases. We train our proposed model on all these subsets and we obtain several insights. Finally, to benchmark human performance, we design human evaluation studies on full-length speech and phrase-level speech clips. Here, listeners are asked to listen each sample *once* and classify real versus fake speech. Subsequently, we compare humans' and models' performance on full length and phrase-level speech clips. In short, we investigate not only the robustness of our proposed model in distinguishing fakeness across TTS systems, but also the comparative performance of machines and humans under varying conditions of speech realism and length.

4. EXPERIMENTS AND RESULTS

For the task of Fake Speech Detection (FSD), we fine-tune LAION-CLAP [33] (`laion/clap-htsat-unfused`), a state-of-the-art audio-text representation model. CLAP is trained on paired audio-text data to produce embeddings that capture both semantic and acoustic characteristics. We append a classification head and train using cross-entropy loss. We train separate models for each experiment as highlighted in Section 3. For full-length data finetuning, we use Adam optimizer [34] (learning rate 5×10^{-5}), a batch size of 64, and train for 30 epochs. We evaluate detection accuracy on held-out test sets across multiple scenarios: (1) Full-length cross-evaluation where each of our model trained with fake speech data from one TTS and tested on others; (2) human listener performance on full-length speech and phrase-level speech clips; (3) phrase-level cross-evaluation of model; and (4) a combined-model trained on both full-length TTS and AV-DeepFake data.

4.1. Cross Evaluation of our Full-Length FSD Models

Table 1 summarizes the detection accuracies of full-length fake speech. The columns represent the subset of our test dataset (each denotes the TTS model used to generate fake speech), while AV-DeepFake denotes data from the AV-DeepFake1M dataset [4]. The rows of tha table depicts subset (generated from different TTS models) of train dataset on which our model is fine-tuned on and then evaluated across all test subsets. Finally, we also report the performance of the single *combined model* trained on the union of all synthetic sources *i.e.* combining fake speech generated from all TTS models and AV-DeepFake samples. We also report on a zero-shot method, VideoLLaMA2. We observe that our models trained on test data from all TTS systems (OpenVoice, XTTS, YourTTS) obtain high self-test accuracies but fail to generalize to AV-DeepFake (chance performance). The AV-DeepFake-trained model performs at chance on the modern TTS sets. This is because of the distribution shift between train and test split. To tackle this, we train our model on test split combining subsets from all TTS and AV-DeepFake data, which achieved robust detection across all test splits as tabulated in Table 1. Surprisingly, VideoLLaMA2 performs at chance (50%) on all test splits, indicating that current Multimodal-LLM models are not able to identify fakeness given audio data. Additionally, from Table 1 we see that our model trained on train split with *OpenVoice* full-length speech performs consistently well in cross-evaluation on other TTS-based test-splits. So, in this work, we have decided to further perform all ablation studies using train-test split based from OpenVoice TTS model [8].

Table 1. Full-length Fake Speech Detection accuracy (%) for cross-system evaluation. We train our model on our train dataset splits (fakes) generated by different TTS models and test on test splits generated by same TTS models. Ours (combined) indicates combining all TTS splits and AV-DeepFake data.

Model / Train Set	OpenVoice	XTTS	YourTTS	AV-DeepFake
VideoLLaMA2 (zero-shot)	50.00%	50.00%	50.00%	50.00%
Ours (OpenVoice)	**97.72%**	95.95%	**98.57%**	50.00%
Ours (XTTS)	67.62%	91.59%	91.72%	50.00%
Ours (YourTTS)	59.18%	60.48%	99.96%	50.00%
AV-DeepFake	50.00%	50.00%	50.00%	50.00%
Ours (Combined)	76.31%	**96.85%**	97.20%	**79.11%**

4.2. Human Evaluation (Full-Length and Phrase Level)

We compute human performance of FSD on both full-length and phrase-level (*OpenVoice*) speech test splits. Eight human listeners participated in our evaluation. They classified 50 real and 50 fake samples from each of the test splits. Table 2 reports the aggregated human performance in tasks at the full-length and phrase-level speech. Humans reach an average accuracy of 75.07% on full-length speech, while their performance on phrase-level speech clips improve with increase in phrase length of the given speech. We believe this is because the current TTS models cannot yet accurately capture the intonation, rhythm and other features of real speech while generating longer audios which helps humans to recognize these fakes easily. We also plotted this trend of human performance in Figure 3.

4.3. Phrase-Level Cross-Evaluation

We cross-evaluate fake speech detectors at phrase level in Table 3. This shows the cross-evaluation accuracies of our model trained on phrase speech of varying lengths generated from *OpenVoice*-TTS, followed by testing on phrase-level test splits from OpenVoice, XTTS, and YourTTS of corresponding lengths that it is trained on. We note that the FSD per-

Table 2. Human listener Accuracy (%) on full-length speech and *OpenVoice* generated phrase clips of our test split. Human performance increases with increase in length of speech.

Setting / Phrase Length	Accuracy (%)↑
Full-length	75.07%
1-word	27.00%
2-word	57.50%
4-word	52.00%
8-word	83.50%

Table 3. Phrase-level cross-evaluation Accuracy (%) of our model trained on phrase clips from *OpenVoice*-TTS. Our model performance increases with length of speech.

Phrase Length	OpenVoice	XTTS	YourTTS
1-word	59.56%	51.25%	49.89%
2-word	58.65%	50.02%	50.00%
4-word	87.55%	40.92%	37.50%
8-word	99.57%	49.90%	49.70%

Table 4. The Accuracies (%) of Ours (combined) as in Table 1 (%) on Phrase-level speech on *OpenVoice* clips.

Phrase Length	Accuracy (%) ↑
1-word	78.48%
2-word	85.52%
4-word	89.85%
8-word	80.41%

formance of our model improves remarkably with increase in phrase length on *OpenVoice* generated test split as in Figure 3, whereas accuracies on other TTS systems' data remain near chance. This shows that the models trained on shorter phrase-length speech are not able to generalize well to data generated by other TTS systems. This is in contrast to the performance of our model trained on full-length *OpenVoice* speech data that we see in Table 1. In Figure 3, we compared our model performance with humans. We show that our trained model performs better compared to humans at all phrase-lengths and in full-length speech.

4.4. Combined-Model Phrase Results

We evaluate the *Ours (combined)* of Table 1 on phrase-level test split as in Table 4. This reports the combined model's phrase-level performance, demonstrating robust FSD performance even for shorter phrases. This indicates that the combined model is able to capture subtle artifacts at a fine-grained level, and generalize beyond full-length utterances.

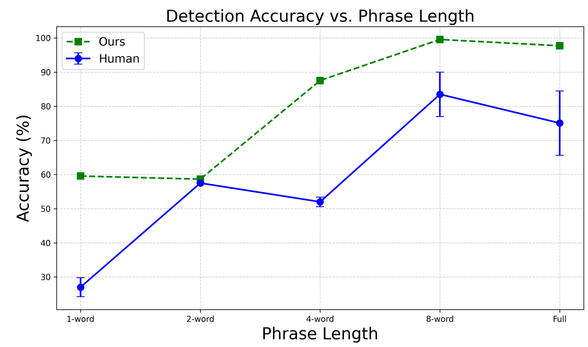

Fig. 3. The above figure shows the Detection accuracy vs. Phrase length for human listeners (with std. deviation) and our proposed model trained on *OpenVoice*-generated phrases and full-length speech. We can see that our trained model outperforms humans on all lengths of phrase-level speech.

5. DISCUSSION

Here are some notable insights from our experiments and ablation studies: a) Our proposed Fake Speech Detector model outperformed the human listeners on full-length speech and phrase-level speech as in Figure 3. b) For very short phrases, human performance is below chance, whereas the model still maintains moderate accuracy, indicating robustness of our model on the length of the given speech, c) Both our model and humans improve as phrases get longer, indicating that more words provide more detectable artifacts, d) our model trained on *OpenVoice*-generated fakes performed the best compared to the models trained on the YourTTS and XTTS, yielding high accuracies across all TTS-based test splits. Fine-tuning of our model on multiple TTS sources i.e. *Ours (combined)* substantially improves robustness. This is evident from our observations in Table 1 and 4 that this model performed well across TTS systems in both full-length and phrase-level and AV-DeepFake data, thus resulting in a model that can efficiently identify fake speech.

6. CONCLUSION

In this paper, we propose a content-agnostic Fake Speech Detection (FSD) model. For this, we curate a large-scale real / fake speech paired dataset and benchmark. We systematically investigate the effects of the length of the input speech sample on our proposed fake detection model. In addition, we also compare the model performance with that of humans for all lengths of given speech samples. We discussed several insights from the above experiments and ablation studies. Our results demonstrate superior performance over recent state-of-the-art FSD models and humans.

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